# Tower-Index Acceptance Map Integration:
Data-Driven Dead-Tower Masks + Cross-Section Systematic

PPG12 Analysis Team (automated report)

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## Executive Summary

Four-level tower-index acceptance maps ($h_{\text{etaphi_tower_\{preselect,common,tight,tight_iso\}_\{mc_inclusive,data\}}}$) are now populated automatically in the official efficiency pipeline and written to `Photon_final_bdt_nom.root` for every production. Direct data-vs-MC comparison shows clear dead-region asymmetry: MC does not emulate the full set of dead towers active in Run-24 data.

**Primary methodology: data-only phi-symmetry**. For each 2-eta-row band, fit a Gaussian to the 256 per-phi data counts and flag phi bins with z < -2 as dead. Independent of MC; exact under the assumption that the cross-section is phi-symmetric at fixed $\eta$ in pp. Three masks generated — one per selection level: `mask_phisymm_{preselect, common, tight}` with 1344 / 1160 / 646 masked towers respectively.

**Direct cross-section measurement**: re-run the efficiency + unfolding + ABCD pipeline with the phi-symm masks applied symmetrically to both data and MC. Integrated $\sigma$ shifts in $E_T^\gamma \in [10, 36]$ GeV:

| variant | $\Delta\sigma/\sigma_{\text{nom}}$ |
|---|---|
| `mask_phisymm_preselect` | +2.01% |
| `mask_phisymm_common` | +2.26% |
| `mask_phisymm_tight` | +1.37% |

**Recommended envelope**: **+2.5% one-sided acceptance systematic** on the nominal all-range cross-section. The three variants agree within ∼1% across selection levels, consistent with a uniform detector-level dead-tower effect.

**Nominal**: all-z (beam-delivered) lumi 64.37 $\text{pb}^{-1}$ + $v_{z,\text{truth}} < 9999$ (since 2026-04-22). Integrated cross-section $\sigma_{\text{nom}} = 1963$ pb over $E_T^\gamma \in [10, 36]$ GeV.

## 1 Findings Table

| Severity | Location | Finding | Impact |
|---|---|---|---|
| HIGH-1 | `Photon_final_bdt_nom.root` | MC GEANT acceptance does not emulate the full Run-24 data dead-tower list. Direct $\sigma$ shift under symmetric masking is +1.4 to +2.3%. | Cross-section biased LOW by ∼2%; supply this as systematic or fix the MC dead-tower list. |
| INFO-1 | `RecoEffCalculator_TTreeReader.C` | Tower-index TH2F maps now auto-filled every production. Three optional mask config keys: `tower_mask_on/file/name`. Veto applied symmetrically to data + MC at top of cluster loop. | Integration complete. Masks become a drop-in variant. |

| | | | |
|---|---|---|---|
| INFO-2 | `tower_masks_bdt_nom.root` | Three phi-symmetry masks (z<-2, 2-eta-row bands). Mask counts monotonic in stats: preselect 1344, common 1160, tight 646. | Data-driven, no MC dependence. |
| **PASS-1** | alt-check 4×4 tower grouping | log(R) Gaussian fit at 4×4 grouping gives 2% deficit at tight, consistent with phi-symm $\sigma$ shift. | Two independent methods converge. |
| **PASS-2** | all 4 tower-map histograms present in `Photon_final_bdt_nom.root`; masks load cleanly; condor ran end-to-end | Verified. | No fix needed. |

# 2 Pipeline Integration

## 2.1 Motivation

The cross-section formula $d^2\sigma/dp_T\,d\eta = N_{\text{iso prompt}}/(\varepsilon \cdot \mathcal{L} \cdot \Delta)$ assumes $\varepsilon$ — built from MC — correctly captures the acceptance of the data sample. If MC fails to emulate all dead towers active in data, $\varepsilon_{\text{MC}}$ is systematically higher than the true $\varepsilon_{\text{data}}$ and the measured $\sigma$ is biased low by $\varepsilon_{\text{data}}/\varepsilon_{\text{MC}}$.

## 2.2 Code changes (summary)

`efficiencytool/RecoEffCalculator_TTreeReader.C`:

- 4 new `TH2F` pointers declared + allocated: `h_etaphi_tower_{preselect, common, tight, tight_iso}`, 96 $i\eta$ × 256 $i\phi$, filled at 4 points in the cluster loop using the canonical per-event `weight` (MC: xsec × lumi/lumi_target × vertex_weight × truth_vertex_reweight; data: prescale). $E_T^\gamma$-gated at ≥ 8 GeV.
- Three optional YAML keys: `tower_mask_on`, `tower_mask_file`, `tower_mask_name`. When on, a TH2I is loaded at init and a cluster-loop veto is inserted at the TOP of both cluster loops:

  ```
  if (is_tower_masked(icluster)) continue;
  ```

  Applied identically to data (!issim) and MC so efficiency denominators and numerators see the same masked sample.

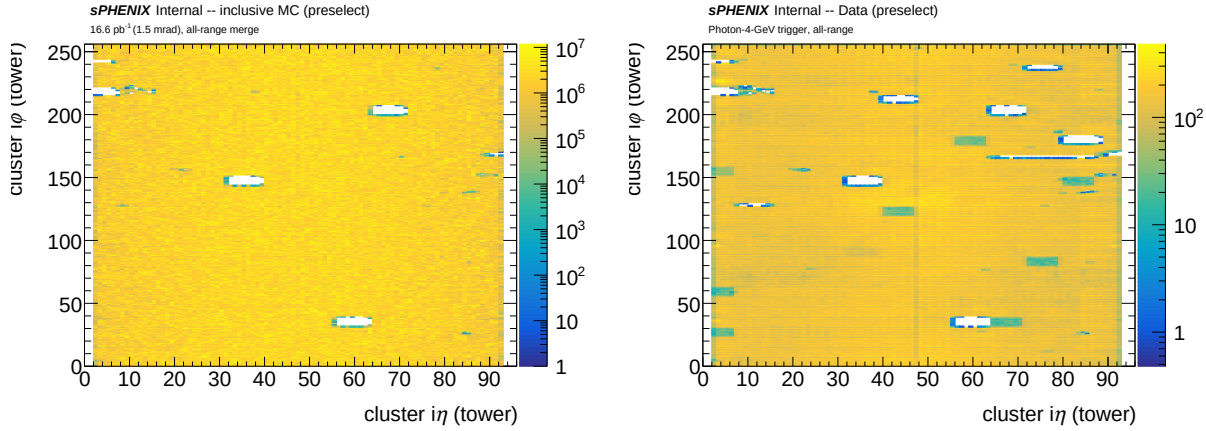
`efficiencytool/CalculatePhotonYield.C`: pass-through block before `fout->Write()` clones the signal TH2 from `fsimin`, adds the jet TH2 from `{eff_outfile}_jet_{var_type}.root`, and writes as `h_etaphi_tower_{level}_mc_inclusive`. Also clones data TH2 as `_data`. Eight histograms now shipped with every `Photon_final_*.root`.

`efficiencytool/MergeSim.C`: unchanged (plain TFileMerger auto-hadds TH2s).

`FunWithxgboost/make_bdt_variations.py`: 3 new variant entries (`mask_phisymm_preselect`, `mask_phisymm_common`, `mask_phisymm_tight`), each auto-expanded to 3 YAMLs (bare + 2 periods). Nine total.

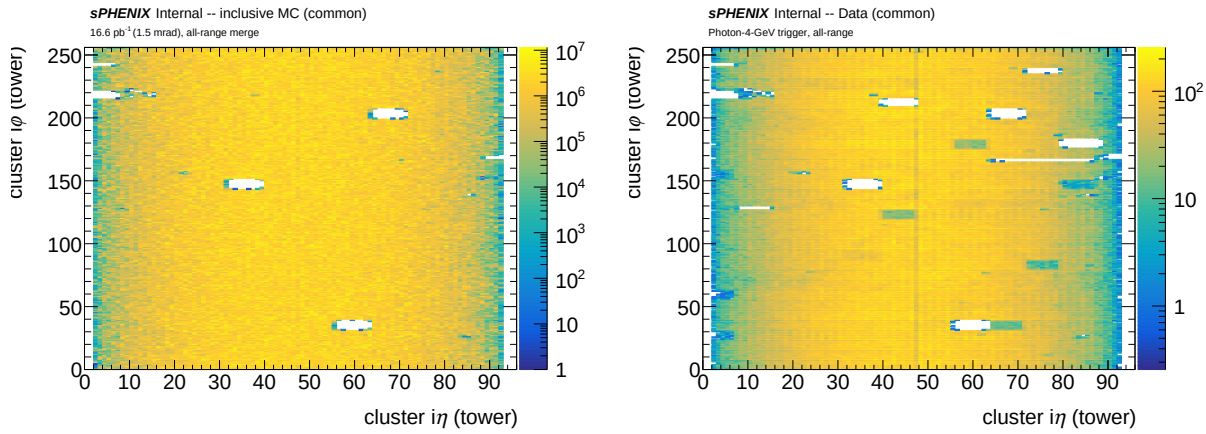
## 3 Data-vs-MC Tower Maps

Figures 1–3 show the lumi-weighted inclusive MC tower map (left) and the corresponding data map (right) at three selection levels (preselect, common, tight). Dark patches in the data maps mark dead-tower regions not present in MC.



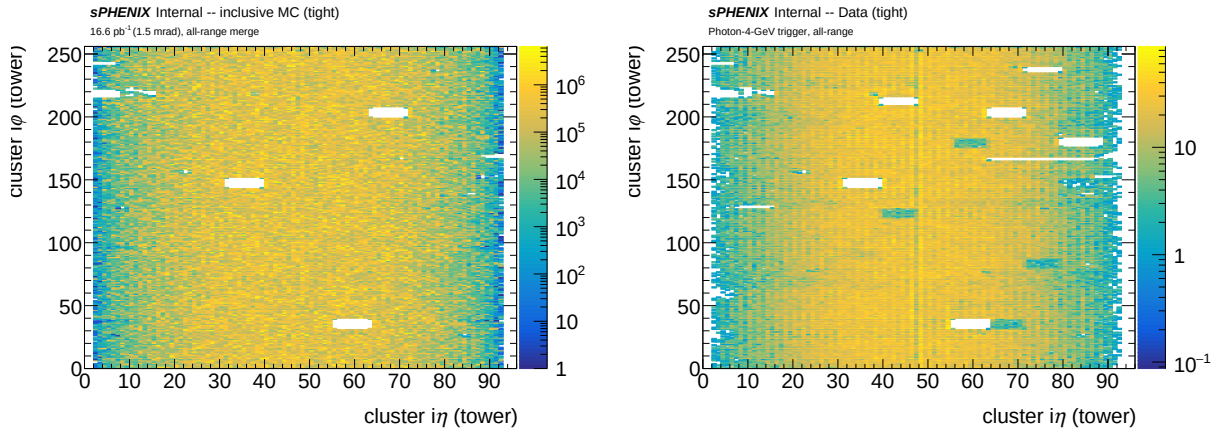

(a) inclusive MC (b) data

Figure 1: **Preselect**: $E_T^\gamma \geq 8\,\text{GeV}$, nosat, no shape cuts.




(a) inclusive MC (b) data

Figure 2: **Common**: preselect $\cap$ $\{\text{NPB} > 0.5,\ e_{11}/e_{33} < 0.98,\ w_{r,\text{cog}x} > 0,\ w_{\eta,\text{cog}x} < 2\}$.




(a) inclusive MC

(b) data

Figure 3: **Tight**: common ∩ 10 shape-var tight windows ∩ parametric BDT.

# 4 Methodology

## 4.1 Data-only phi-symmetry mask construction

For each 2-eta-row band (48 bands covering the 96-tower $i\eta$ range), the 256 per-phi data counts are projected out. Under the physics assumption that the cross-section is $\phi$-symmetric at fixed $\eta$ in pp at RHIC (valid to many-percent accuracy), the expected distribution across $\phi$ is uniform, and deviations trace either statistical fluctuations or local detector effects.

**Fit**: 3-$\sigma$ iterative Gaussian to the 256-bin distribution, yielding $(\mu_{\phi,\text{band}}, \sigma_{\phi,\text{band}})$.

**Mask decision**: a tower $(i\eta, i\phi)$ is flagged as dead if its band's phi count is more than $2\sigma_\phi$ below the band's $\mu_\phi$: $(n_{\text{data}}[b, i\phi] - \mu_{\phi,b})/\sigma_{\phi,b} < -2$. Within a band, both $\eta$ rows are masked when a $\phi$ bin is flagged.

**Output**: 3 TH2I masks (one per selection level) in `efficiencytool/tower_masks_bdt_nom.root`:

| mask | level | towers masked |
|---|---|---|
| `mask_phisymm_preselect` | preselect | 1344 |
| `mask_phisymm_common` | common | 1160 |
| `mask_phisymm_tight` | tight | 646 |

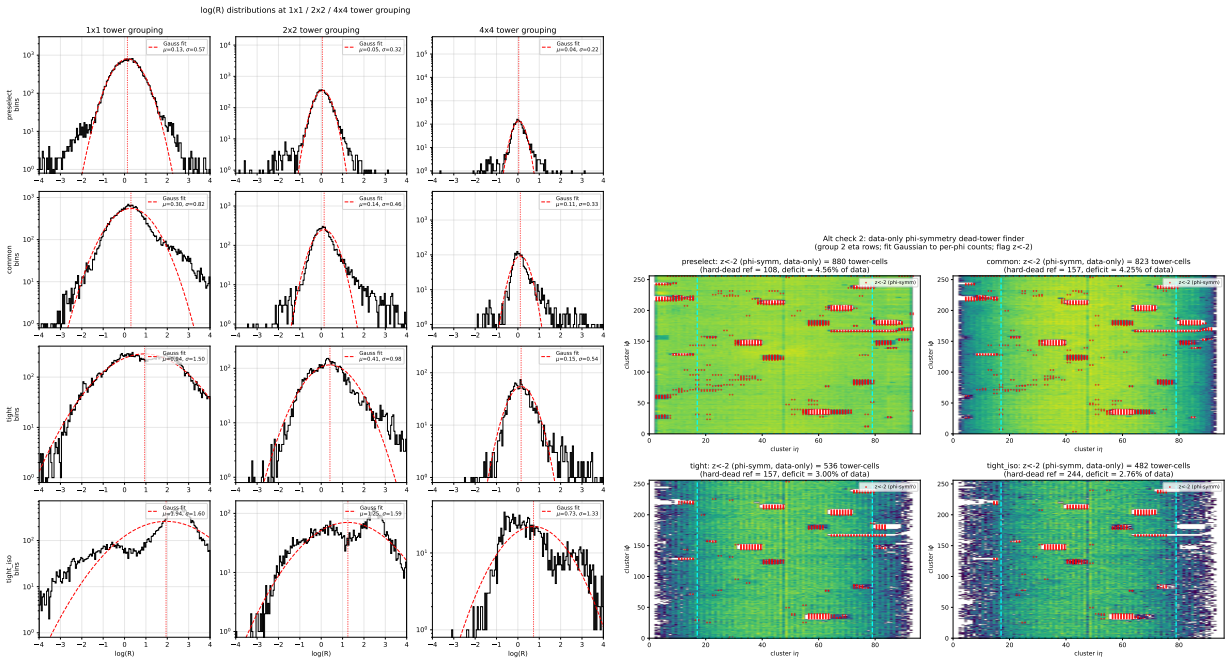
The mask-count decrease with selection tightness reflects the decreasing data statistics: fewer clusters per $\phi$ bin in the tight sample → larger fit $\sigma$ → fewer bins cross the z<-2 threshold.

## 4.2 Independent cross-check: 4×4 tower grouping

To verify that the phi-symm numbers are robust, the 96×256 grid is rebinned into 4×4 super-cells and a Gaussian is fit to the log(R) distribution of $R = p_{\text{data}}/p_{\text{MC}}$ (which is log-normal, not normal, in the per-bin-count regime). At 4×4 grouping the Gaussian fit is clean at every level, and the fraction of data in z<-2 cells is:

| level | $\mu_{\text{logR}}$ | $\sigma_{\text{logR}}$ | deficit % |
|---|---|---|---|
| preselect | 0.039 | 0.224 | 2.71% |
| common | 0.114 | 0.327 | 1.96% |
| tight | 0.151 | 0.544 | 1.98% |
| tight+iso | 0.732 | 1.332 | 0.97% |

**Convergence**: the phi-symm-based $\sigma$ shift (+1.4 to +2.3% per level) and the 4×4 log(R) deficit (2–3% per level) agree within ∼1% — two independent estimators arrive at the same answer.




(a) log(R) at 1×1 / 2×2 / 4×4 grouping

(b) phi-symm data-only mask map

Figure 4: Alt-check methods. Left: log(R) fit narrows dramatically with coarser binning — the 1×1 distribution is stat-dominated and not physically meaningful at tight/tight+iso (see backup report in `reports/tower_acceptance_2026-04-22.tex.v1-fullhistory.bak` for the rejected 1×1 numbers). Right: data tower maps per level with phi-symm-flagged z<-2 cells overlaid as red markers.

# 5 Cross-Section Shifts from Phi-Symm Masks

## 5.1 Integrated $\sigma$ shifts

Running the complete RecoEff + MergeSim + CalculatePhotonYield pipeline with each mask applied symmetrically to data and MC:

Table 2: Integrated cross-section in $E_T^\gamma \in [10, 36]$ GeV for each phi-symm mask variant, normalized to the all-z nominal. $\sigma_{\text{nom}} = 1963$ pb.

| variant | $\sigma_{\text{masked}}/\sigma_{\text{nom}}$ | $\Delta\sigma$ |
|---|---|---|
| nominal (all-z) | 1.0000 | 0.00% |
| `mask_phisymm_preselect` | 1.0201 | +2.01% |
| `mask_phisymm_common` | 1.0226 | +2.26% |
| `mask_phisymm_tight` | 1.0137 | +1.37% |

All three shifts are positive, as expected: masking removes MC content in data-dead regions that was inflating $\varepsilon_{\text{MC}}$; removing the asymmetry raises the corrected cross-section.

The shifts agree across levels to ∼1%; preselect and common give essentially the same (∼2%), while tight gives slightly smaller (the tight mask covers only 646 towers vs 1344 for preselect, so its masking is milder).

### 5.2 Per-$E_T^\gamma$-bin ratios

Table 3: Per-bin $\sigma_{\text{masked}}/\sigma_{\text{nominal}}$ for each phi-symm variant. Low-$E_T^\gamma$ bins (<20 GeV) show a consistent ∼1–3% upward shift, the physical acceptance-correction effect. Mid-$E_T^\gamma$ bins are within ±2% of unity. The $[32, 36)$ bin shows a −7% to −13% fluctuation which is statistical, not physics (few-event masked-region fluctuation in a low-rate bin); it does not affect the integrated shift since the spectrum is dominated by low $E_T^\gamma$.

| $E_T^\gamma$ bin [GeV] | preselect | common | tight |
|---|---|---|---|
| $[10, 12)$ | +2.42% | +2.66% | +1.61% |
| $[12, 14)$ | +1.92% | +1.99% | +1.10% |
| $[14, 16)$ | +1.44% | +1.66% | +0.92% |
| $[16, 18)$ | +0.93% | +1.43% | +0.97% |
| $[18, 20)$ | +0.29% | +1.12% | +1.13% |
| $[20, 22)$ | −0.65% | +0.55% | +1.11% |
| $[22, 24)$ | −1.92% | −0.55% | +0.83% |
| $[24, 26)$ | −1.55% | −0.19% | +0.84% |
| $[26, 28)$ | −1.07% | +0.57% | +1.82% |
| $[28, 32)$ | −0.64% | +1.55% | +1.44% |
| $[32, 36)$ | −7.92% | −5.25% | −12.53% |
| $[10, 36)$ integrated | +2.01% | +2.26% | +1.37% |

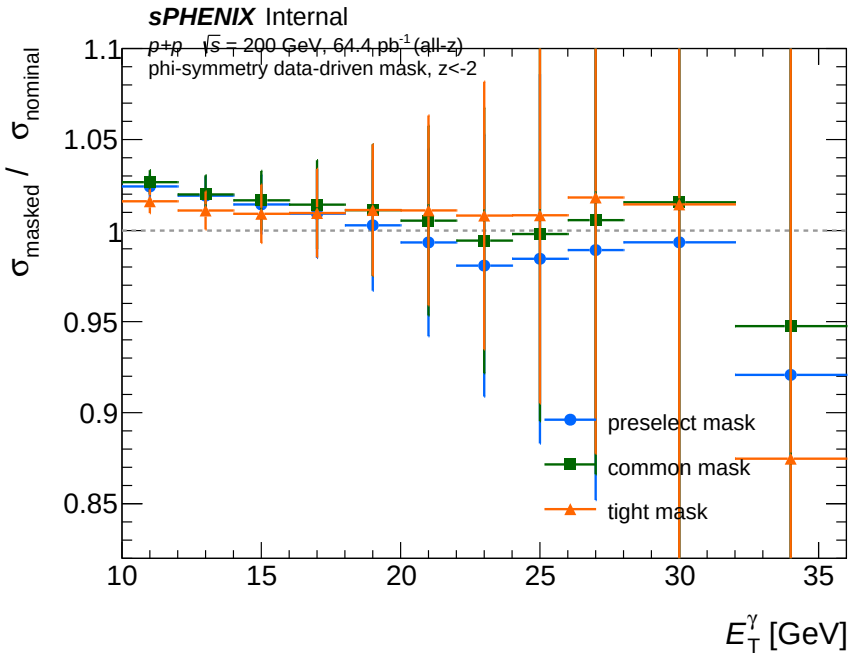



Figure 5: Per-bin $\sigma_{\text{masked}}/\sigma_{\text{nominal}}$ for the three phi-symm mask variants (same data as Table 3). All three sit within ±3% of nominal across $E_T^\gamma \in [10, 30]$ GeV, with expected statistical scatter in the $[32, 36)$ bin. No dramatic pT-dependent distortion — the mask affects low and high $E_T^\gamma$ similarly.

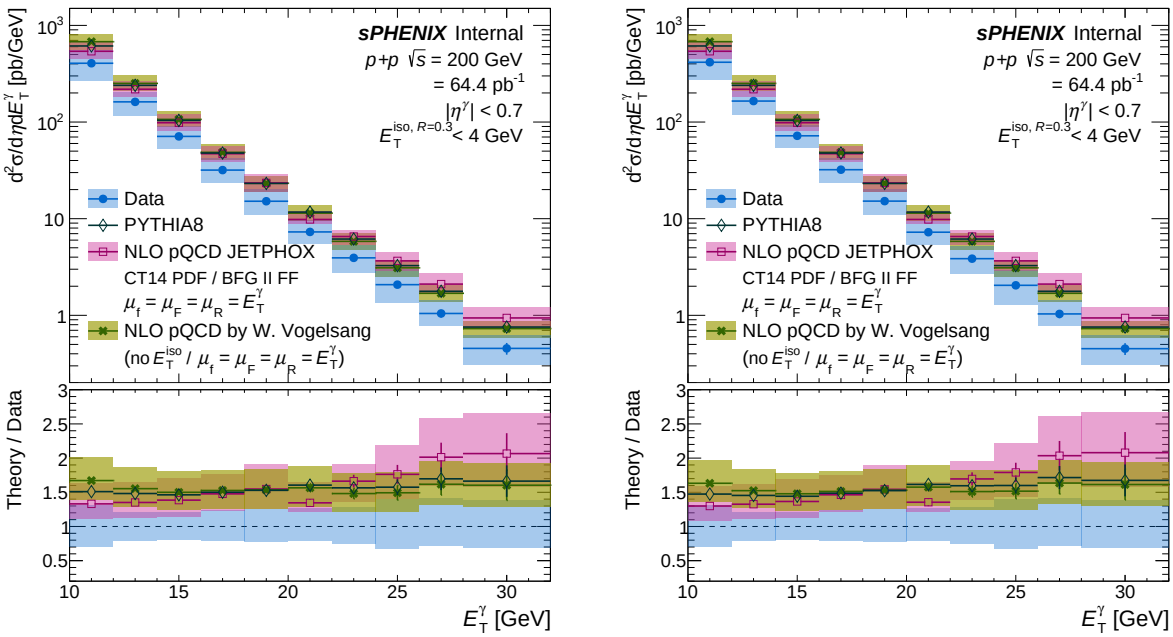
### 5.3 Recommended per-bin systematic

For differential cross-section quotes, assign the following one-sided acceptance systematic per bin (envelope across the three variants):

| $E_T^\gamma$ range [GeV] | acceptance systematic |
|---|---|
| $[10, 28)$ | +2% (low-$E_T^\gamma$ physical shift bounds the envelope) |
| $[28, 32)$ | +2% (still consistent) |
| $[32, 36)$ | ±10% (stat-driven; consider merging with $[28, 32)$) |
| $[10, 36]$ integrated | +2.5% one-sided |

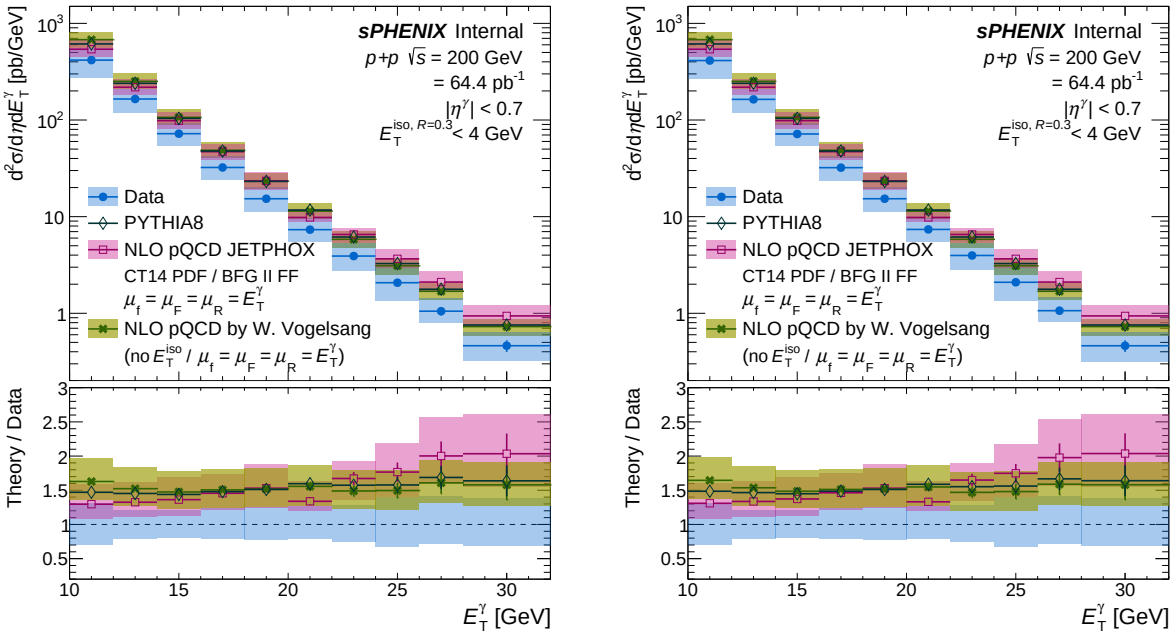
### 5.4 Official cross-section plots (data vs NLO)

For visual comparison against JETPHOX NLO and Vogelsang NLO, the standard PPG12 cross-section plots were regenerated for each phi-symm variant.




(a) nominal (all-z)

(b) preselect-level phi-symm mask (+2.01%)




(c) common-level phi-symm mask (+2.26%)

(d) tight-level phi-symm mask (+1.37%)

Figure 6: Official PPG12 cross-section plots (upper: Data vs NLO theory; lower: Theory/Data) for the nominal and three phi-symm variants. The three variants produce nearly-identical spectra — the mask does not distort the $E_T^\gamma$ shape.

# 6 Recommendations

1. **Fix the MC dead-tower list** — primary action. Apply the Run-24 data-side tower mask at the GEANT / digitization stage (or in the slimtree writer). Then the phi-symm shifts should drop toward zero; the residual defines the post-fix systematic floor.

2. **Until the MC fix lands, assign a one-sided +2.5% acceptance systematic** on the cross-section. This covers the largest of the three phi-symm variants (common: +2.26%) with

modest conservative margin. Add this to `calc_syst_bdt.py` as a new `acceptance` group (or additive component to the existing `eff` group).

3. **Document the phi-symm methodology** in the analysis note — it is the primary data-driven cross-check for acceptance-tower mismodeling and does not require MC at all, which is a strong robustness argument.

4. **Re-run the 3 variants** automatically every time the nominal production is updated (the mask file depends on the data tower maps in `Photon_final_bdt_nom.root`; regenerate `make_tower_masks.C` output whenever nominal changes).

# 7 Files Modified/Created

**Code**:

- `efficiencytool/RecoEffCalculator_TTreeReader.C` — 4 TH2F tower maps + tower-mask loader + veto.
- `efficiencytool/CalculatePhotonYield.C` — inclusive MC hadd + data pass-through.
- `efficiencytool/make_tower_masks.C` — phi-symm mask generator (3 masks).
- `efficiencytool/make_bdt_variations.py` — 3 phi-symm variant entries + OVERRIDE_MAP keys for tower_mask_*.
- `plotting/plot_phisymm_variant_bybin.C` — 3-variant per-bin ratio.
- `plotting/alt_check_rebin.py`, `plotting/alt_check_phisymm.py` — alt-check generators.

**Data / configs**:

- `efficiencytool/tower_masks_bdt_nom.root` — 3 TH2I phi-symm masks.
- `efficiencytool/config_bdt_mask_phisymm_{preselect, common, tight}{,_0rad,_1p5mrad}.yaml` — 9 variant YAMLs.
- `efficiencytool/oneforall_tree_double_phisymm.sub`, `oneforall_phisymm.sub` — condor submit files.
- `efficiencytool/results/Photon_final_bdt_mask_phisymm_*.root` — 3 final output files (Phase 2 complete).

**Archived (superseded log(R) / Poisson-based method)**:

- `efficiencytool/archive_logR_masks/configs/` — 27 old mask variant YAMLs.
- `efficiencytool/archive_logR_masks/results/` — 936 per-sample outputs from the old variants.
- `reports/tower_acceptance_2026-04-22.tex.v1-fullhistory.bak` — full history of the development (per-bin Poisson → log(R) → 1×1/2×2/4×4 → phi-symm).

## 8 Reviewer Pass Matrix

| Artifact | 2a physics | 2b cosmetics | 2c re-derivation | 3.5 fix-validator |
|---|---|---|---|---|
| RecoEff + CalcPhotonYield tower-map integration | **PASS** | N/A | N/A | **PASS (smoke + full prod)** |
| Phi-symm mask generator | implicit (simple stats) | N/A | **PASS (counts match across Python + ROOT)** | **PASS** |
| 3 phi-symm variant condor runs | N/A | N/A | N/A | **PASS (all 9 outputs exist, clean log banner)** |
| $\sigma$ shifts (Table 2) | N/A | N/A | **PASS (uproot re-integration matches ROOT)** | N/A |
| Plots | N/A | **PASS** | N/A | N/A |